Assignment 2

1. Grade Checker

Code:

```
# Grade Checker Program

score = int(input("Enter your score: "))

if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
elif score >= 60:
    print("Grade: D")
else:
    print("Grade: F")
```

Answer :

```
Enter your score: 70
Grade: C
>>>
```

Explanation:

The program takes input using input() function.

Converts input into integer using int().

Uses if-elif-else ladder:

If score ≥ 90 → A

Else if ≥ 80 → B

Else if ≥ 70 → C

Else if ≥ 60 → D

Else → F

2. Student Grades

```python
students = {}

while True:
    print("\n1. Add Student")
    print("2. Update Student Grade")
    print("3. Display All Students")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == '1':
        name = input("Enter student name: ")
        grade = input("Enter grade: ")
        students[name] = grade
        print("Student added successfully!")

    elif choice == '2':
        name = input("Enter student name to update: ")
        if name in students:
            grade = input("Enter new grade: ")
            students[name] = grade
            print("Grade updated successfully!")
        else:
            print("Student not found!")

    elif choice == '3':
        if students:
            print("\nStudent Grades:")
            for name, grade in students.items():
                print(name, ":", grade)
        else:
            print("No records found!")

    elif choice == '4':
        print("Exiting program...")
        break

    else:
        print("Invalid choice! Please try again.")
```

Solution:

```
---------------- RESTART: /Users/appus/D

1. Add Student
2. Update Student Grade
3. Display All Students
4. Exit
Enter your choice: 1
Enter student name: Koushek
Enter grade: A
Student added successfully!

1. Add Student
2. Update Student Grade
3. Display All Students
4. Exit
Enter your choice: Krishna
Invalid choice! Please try again.

1. Add Student
2. Update Student Grade
3. Display All Students
4. Exit
Enter your choice: 3

Student Grades:
Koushek : A

1. Add Student
2. Update Student Grade
3. Display All Students
4. Exit
Enter your choice:
```

Explanation:

A dictionary (students) is used to store student data:

Key → Student Name

Value → Grade

The program uses a while loop to repeatedly show options.

If-elif-else statements handle user choices:

Option 1 → Add student

Option 2 → Update grade (checks if student exists)

Option 3 → Display all records

Option 4 → Exit program

Dictionary operations used:

Add → students[name] = grade

Update → same assignment

Check existence → if name in students

3. Write to a File

```
content = input("Enter content to write into the file: ")

with open("output.txt", "w") as file:
    file.write(content)

print("Content written successfully to output.txt")
```

Solution:

```
Enter content to write into the file: rython.py
Content written successfully to output.txt
```

Explanation:

input() is used to take text from the user

open("output.txt", "w"):

Opens the file in write mode

Creates the file if it doesn't exist

Overwrites the file if it already exists

write() function writes the content into the file

close() is used to properly close the file

4. Read from a File

```
file = open("output.txt", "r")
content = file.read()
print("File Content:")
print(content)
file.close()
```

Solution:

```
File Content:
rython.py
>>>
```

Explanation:

open("output.txt", "r"):

Opens the file in read mode

File must exist, otherwise it will show an error

read():

Reads the entire content of the file

print():

Displays the content on the console

close():

Closes the file after reading